

Certified Valuation of Arithmetic Asian Options via Common Gaussian Smoothing

Computable Error Bounds for Projected Euler, Joint Weak Expansions, and Posterior Quantile
Transfer

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Abstract

We develop finite error certificates for the bounded arithmetic Asian call spread $\psi(A) = (A - 95)^+ - (A - 110)^+$, with twelve monthly fixings, in a specified Heston model and its original positive-part variance Euler scheme. A common Gaussian factor gives a nonnegative payoff remainder controlled by a weighted second moment. Validated transform arithmetic includes projection corrections, integration tails, inversion error, complex-logarithm branches, and rounding. At $\theta_* = (3, .045, .23, -.55, .045)$ and $h = 1/768$, the signed Euler-minus-continuous price difference lies in

$$[-0.011024692273, 0.010642371599],$$

and the continuous-model call-spread price lies in

$$[6.508371733, 6.518868974].$$

A second initial-variance point has a complete certificate. The step-grid comparison identifies a remainder-dominated width plateau, and we derive the optimal common geometric scale for the exact nonlinear contribution. Two auxiliary results have separate domains: a common first-order expansion formula with explicit remainder bounds for nine puts and the Asian spread when $\xi = 0$, and posterior target quantile displacement bounds on a restricted continuous prior with $\xi \in [10^{-7}, 10^{-6}]$. The contribution is the weighted payoff structure and complete finite error accounting for the specified original discrete kernel, supported by an executable numerical package.

Keywords: Arithmetic Asian options; Heston model; projected Euler; conditional Gaussian smoothing; computable error bounds; weak error expansions; posterior quantiles

1 Introduction

Numerical valuation of derivatives involves two distinct questions: computing a model price, and proving how closely the discrete model used in the computation approximates the continuous model. For arithmetic Asian options with fixed observation dates, the positive-part payoff, stochastic variance, and projection of the discrete variance are all present. A practically useful conclusion should provide an error in price units at the specified step size and account for payoff conversion, transform evaluation, infinite tails, and rounding within a single error budget.

We organize the theory around this requirement. We study the risk-neutral Heston model and a precisely specified scheme that applies a positive-part projection to the variance Euler update and uses the current variance in the log-price update. The object of interest is the difference between expectations of the actual arithmetic-average payoff. The method begins at the payoff level: identify a Gaussian random factor shared by all observed prices, integrate it exactly, and use the structure of the remaining variables to bound the error. The key quantity is not a negative moment of the variance at an individual time, but

$$E\left[(A - cG)^2 I^{-1/2}\right],$$

where A is the arithmetic average, G is the geometric average, and I is the integrated variance over the first observation interval.

Conditioning and geometric reference variables have a well-established role in Asian option pricing [1, 2, 3, 4]. In particular, the framework of Fusai and Kyriakou covers the Heston model and includes error upper bounds based on conditional variances [4]. The affine structure of Heston's model supplies the transform machinery [5], while discrete geometric Asian options also admit specialized recursions and publicly available implementations [6]. Building on these foundations, we carry out three specific tasks.

First, we formulate a structural condition $\mathbf{H}$ that can be checked independently. Conditional Gaussian smoothing yields a nonnegative convexity remainder with explicit constants; combining this remainder with a certificate for finitely many transforms gives an enclosure of the true price difference. The condition applies both to lognormal models with deterministic variance and to the stochastic-variance model and actual projected Euler kernel verified here.

Second, we account for every computable error contribution for the specified original scheme. The quadratic remainder of the arithmetic payoff is controlled by 91 real loadings and a nonnegative Laplace integral; the linear part is computed from thirteen complex loadings. The result is a complete one-year price certificate at the original parameter point, rather than an asymptotic order with an unknown constant.

Third, we develop two further mathematical applications: a joint weak expansion for ten payoffs on a deterministic-variance reference family, and control of the true target quantiles under a continuous prior with small positive volatility of volatility, using likelihood enclosures over the full parameter cells and a joint posterior coupling. Each application has explicit assumptions and quantitative conclusions.

The logical structure of the paper is

$$\left. \begin{array}{l} \text{A : deterministic positive integrated variance} \\ \text{C : verifiable Heston/projected Euler conditions} \end{array} \right\} \Longrightarrow \text{H}, \quad (1.1)$$

$$\text{H} \xRightarrow{\Phi} \text{M} \Longrightarrow \text{B}.$$

Here Φ denotes conditional Gaussian smoothing and rigorous transform error accounting; M consists of the payoff decomposition with explicit constants and transform error bounds; and B is a rigorous enclosure of the actual price difference. Section 2 defines these objects precisely, and Sections 3--5 prove the main chain of implications. Sections 6 and 7 develop the weak expansion and posterior transfer, respectively. Section 8 presents the numerical certificates and their financial use, and Section 9 specifies the domain of validity.

1.1 Comparison with conditional Asian bounds

Fusai and Kyriakou [4] already obtain optimized conditioning bounds for Heston Asian options (Sections 2.2--3), conditional-variance error bounds (Theorem 4, equations (51)--(54)), and experiments at several monitoring frequencies (Section 5.1, Table 5). The following table identifies the particular mathematical and computational objects studied here. This comparison concerns that reviewed paper; it does not establish global priority. Our principal result is the finite error enclosure for the original kernel (2.2), built from a payoff remainder weighted by first-period integrated variance and a validated computation of every stated error contribution.

Table 1. Comparison with the conditional Asian bounds of Fusai and Kyriakou.

Object	Fusai--Kyriakou (reviewed source)	Present result
Payoff approximation	Optimized conditional lower bound and conditional-variance error (Theorem 4, p. 15).	Exact linear-reference decomposition with nonnegative remainder controlled by a weighted second moment (Lemma 3.1).
Discrete numerical law	QE simulation benchmark (Appendix B, p. 22); no certificate for kernel (2.2) is reported.	Projection correction between an algebraic transform recursion and the specified original positive-part variance Euler law, using current variance in the stock update (Appendix B).
Transform computation	Fourier conditional moments (equations (55)–(56), pp. 15–16) and FFT implementation (Section 5, pp. 16–17).	Finite real and complex loading catalogs, certified quadrature and infinite-tail bounds, and complex-logarithm branch checks (Appendices A–C).
Finite arithmetic	An outward-rounded error budget is not reported in the reviewed source.	Rational endpoint combination and ball-arithmetic enclosures include rounding with the other finite errors (Appendices A–C and G).
Certified output	Lower-bound approximation error (Theorem 4); no executable signed Euler-minus-continuous enclosure is reported.	Signed $p_h - p_c$ interval at the specified parameters and grid (Theorem 4.1 and Corollary 4.2), traceable to the accompanying executable package.

The increment studied here is the weighted remainder and its combination with error control for the original projected kernel. Section 4 analyzes the resulting certificate width.

1.2 Three distinct domains of validity

The contract in all monetary Asian conclusions is the arithmetic call spread in (2.3), whose payoff is bounded by 15. Table 2 distinguishes the domains of the three results. The posterior example concerns a near-deterministic perturbation regime rather than calibration stability at the main Heston point.

Table 2. Domains of the three certified results.

Result	Domain and interpretation
Main price certificate	The point (4.3) and the separate $v_0 = .04$ point in Section 8.3, $\xi = .23$, twelve monthly fixings, original projected Euler, $h = 1/768$.
Ten-payoff weak expansion	Theorem 6.1 under (6.1); numerical family (6.10): $\xi = 0$, $\kappa = 3$, $\bar{v} = .045$, $v_0 \in [.03, .06]$, twelve monthly fixings, aligned $h \leq 1/768$.
Posterior quantile transfer	Independent uniform prior (7.5), $\xi \in [10^{-7}, 10^{-6}]$, exact synthetic quotes in Appendix F.

2 Models and structural conditions

2.1 The continuous model, discrete kernel, and payoff

Let W, B be independent standard Brownian motions, and let $\theta = (\kappa, \bar{v}, \xi, \rho, v_0)$. Assume $S_0, \kappa, \bar{v}, v_0 > 0$, $\xi \geq 0$, $|\rho| < 1$, and $r \in \mathbb{R}$. The discrete model uses $h > 0$, with each t_i/h a positive integer. The continuous model is

$$\begin{aligned}
 dV_t &= \kappa(\bar{v} - V_t) dt + \xi \sqrt{V_t} dW_t, \\
 dZ_t &= (r - V_t/2) dt + \sqrt{V_t}(\rho dW_t + \sqrt{1 - \rho^2} dB_t), \\
 S_t &= S_0 e^{Z_t}, \quad Z_0 = 0, \quad V_0 = v_0.
 \end{aligned} \tag{2.1}$$

On the aligned grid with step size h , define the original discrete kernel Q_h by

$$\begin{aligned}
 V_{j+1}^h &= \left[(1 - \kappa h) V_j^h + \kappa \bar{v} h + \xi \sqrt{h V_j^h} G_j \right]^+, \\
 Z_{j+1}^h &= Z_j^h + (r - V_j^h/2)h + \sqrt{h V_j^h}(\rho G_j + \sqrt{1 - \rho^2} H_j),
 \end{aligned} \tag{2.2}$$

where G_j, H_j are independent standard normal variables, the two updates at the same step share G_j , and the initial conditions agree. In particular, the price update uses V_j^h .

Fix $0 < t_1 < \dots < t_n = T$, and set

$$A = \frac{1}{n} \sum_{i=1}^n S_{t_i}, \quad G = \left(\prod_{i=1}^n S_{t_i} \right)^{1/n}, \quad \psi(x) = (x - K_1)^+ - (x - K_2)^+, \tag{2.3}$$

where $0 < K_1 < K_2$. Write

$$p_c = e^{-rT} E_P \psi(A), \quad p_h = e^{-rT} E_{Q_h} \psi(A), \quad e_h = p_h - p_c. \tag{2.4}$$

This sign convention for the error is used throughout. The numerical example uses $n = 12, t_i = i/12, T = 1$, and

$$S_0 = 100, \quad r = .01, \quad K_1 = 95, \quad K_2 = 110, \quad h = 1/768. \tag{2.5}$$

The calibration payoffs are the nine European puts with maturities $1/4, 1/2, 1$ and strikes $90, 100, 110$.

2.2 The common Gaussian structure H_s

A pricing law ν is said to satisfy $H_s(c)$ if there exist a sub- σ -algebra $\mathcal{G}$, positive $\mathcal{G}$ -measurable random variables $a_1, \dots, a_n, \sigma$, and a random variable U such that

$$U \mid \mathcal{G} \sim N(0, \sigma^2), \quad S_{t_i} = e^U a_i. \quad (2.6)$$

Set $a = n^{-1} \sum_i a_i$ and $g = (\prod_i a_i)^{1/n}$. We require

$$E_\nu(A + G) < \infty, \quad D_\nu(c) := E_\nu \left[\frac{(A - cG)^2}{\sigma} \right] < \infty. \quad (2.7)$$

Here $c > 0$ is a constant chosen in advance. Condition (2.6) describes the stochastic structure of the model, while (2.7) is an integrability condition that can be checked through moments or Laplace transforms. Neither condition already contains the price-difference conclusion to be proved.

2.3 The finite verification condition H_v

The structural condition must also be connected to finite computation. The condition H_v used here consists of the following inputs:

1. A computable upper bound for $D_\nu(c)$, obtained from enclosures at finitely many Laplace nodes, provable quadrature enclosures, and a bound on the infinite tail.
2. Outward enclosures for the finite set of transform loadings required in Section 3. For the discrete kernel, each loading also has a computable bound on the difference between the true kernel and its algebraic recursion.
3. Complete bounds on the infinite-frequency tail and periodization error in Fourier inversion, together with branch control for the continuous complex transform.

These inputs are defined in terms of specific transforms or moments. Their connection to the target price is established by the theorems below. We write $H = H_s + H_v$. The finite catalogs, precision, tail-bound constants, and kernel error formulas are specified in Appendices A--C.

2.4 The intermediate property and target conclusion

The property M has two components: an exact decomposition of the actual payoff into a linear term and a remainder, and signed enclosures for each remainder and each linear-transform error. The target conclusion B is a definite finite interval $[\underline{e}, \bar{e}]$ such that

$$e_h \in [\underline{e}, \bar{e}]. \quad (2.8)$$

Each endpoint is obtained by finitely many rational operations and validated enclosures of transcendental functions.

3 The core argument: from a common Gaussian factor to a weighted remainder

3.1 A conditional smoothing lemma

Lemma 3.1. Suppose that $H_s(c)$ holds. For $K > 0$, define

$$R_{K,\nu} = E_\nu[(A - K)^+ - (cG - K)^+ - (A - cG)\mathbf{1}_{\{cG > K\}}]. \quad (3.1)$$

Then

$$0 \leq R_{K,\nu} \leq \frac{D_\nu(c)}{2K\sqrt{2\pi}}. \quad (3.2)$$

Proof. Convexity of the positive-part function gives pathwise nonnegativity. Conditional on $\mathcal{G}$, let

$$F_K(x) = E[(xe^U - K)^+ | \mathcal{G}], \quad x > 0.$$

Since $\sigma > 0$, this function is twice differentiable, with

$$F_K''(x) = \frac{K}{\sigma x^2} \varphi\left(\frac{\log(x/K)}{\sigma}\right) \leq \frac{e^{2\sigma^2}}{K\sigma\sqrt{2\pi}}, \quad (3.3)$$

where φ is the standard normal density. The final bound follows by maximizing at $\log(x/K) = -2\sigma^2$.

Applying Taylor's formula with integral remainder between the positive arguments a, cg gives

$$0 \leq F_K(a) - F_K(cg) - (a - cg)F_K'(cg) \leq \frac{e^{2\sigma^2}(a - cg)^2}{2K\sigma\sqrt{2\pi}}.$$

The conditional derivative satisfies

$$(a - cg)F_K'(cg) = E[(A - cG)\mathbf{1}_{\{cG > K\}} | \mathcal{G}].$$

Since

$$E[(A - cG)^2 | \mathcal{G}] = e^{2\sigma^2}(a - cg)^2,$$

taking expectations proves (3.2). The call and linear terms are integrable because $E(A + G) < \infty$. The conditional second-moment identity is used in the sense of nonnegative conditional expectations, and the weighted remainder is controlled by $D_\nu(c) < \infty$. $\square$

The crucial point is that the two factors $e^{2\sigma^2}$ match exactly. Consequently, the curvature contribution from conditional smoothing can be expressed directly as a weighted second moment under the true model. The residual $A - cG$ may have either sign.

3.2 The payoff decomposition

Define

$$\mathcal{L} = \psi(cG) + (A - cG)\mathbf{1}_{\{K_1 < cG \leq K_2\}}. \quad (3.4)$$

The conditional normal variance is strictly positive, so G has no atom at any positive threshold. By (3.1),

$$E_\nu \psi(A) = E_\nu \mathcal{L} + R_{K_1, \nu} - R_{K_2, \nu}. \quad (3.5)$$

This identity retains the original arithmetic payoff as the object of interest, while producing a linear part that can be computed using a small number of transforms.

Let $Y = \log(G/S_0)$ and $y_K = \log(K/(cS_0))$, and define the distribution functions of two positive measures by

$$F_{0, \nu}(y) = \nu(Y \leq y), \quad F_{A, \nu}(y) = E_\nu[A\mathbf{1}_{\{Y \leq y\}}].$$

Expanding (3.4) on each of its intervals gives

$$E_\nu \mathcal{L} = (K_2 - K_1) + K_1 F_{0, \nu}(y_{K_1}) - K_2 F_{0, \nu}(y_{K_2}) - F_{A, \nu}(y_{K_1}) + F_{A, \nu}(y_{K_2}). \quad (3.6)$$

The corresponding transforms are

$$\hat{\mu}_0(z) = E_\nu e^{zY}, \quad \hat{\mu}_A(z) = \frac{S_0}{n} \sum_{i=1}^n E_\nu e^{Z_{t_i} + zY}. \quad (3.7)$$

Thus the linear part requires thirteen loadings when $n = 12$.

4 The main theorem and the complete price certificate

Theorem 4.1 (A signed enclosure of the price difference between two models). Suppose that P, Q_h each satisfy $H_s(c)$, and that computable constants satisfy $D_P(c) \leq d_P, D_Q(c) \leq d_Q$. If H_v provides

$$e^{-rT}(E_Q \mathcal{L} - E_P \mathcal{L}) \in [\ell, u], \quad (4.1)$$

then, with $\gamma = e^{-rT}/(2\sqrt{2\pi})$,

$$e_h \in \left[\ell - \gamma \left(\frac{d_Q}{K_2} + \frac{d_P}{K_1} \right), \quad u + \gamma \left(\frac{d_Q}{K_1} + \frac{d_P}{K_2} \right) \right]. \quad (4.2)$$

Proof. Apply (3.5) to each law and subtract. The lower-end contributions are $-R_{K_2, Q} - R_{K_1, P}$, and the upper-end contributions are $R_{K_1, Q} + R_{K_2, P}$. Lemma 3.1 and (4.1) give the result. $\square$

The final step of the main theorem consists only of combining intervals. Its mathematical substance lies in Lemma 3.1 and the rigorous verification of H_v . The resulting interval is generally not symmetric about zero, and its center is retained in the computation.

Corollary 4.2 (The specified Heston parameter point). For (2.1)–(2.5), take

$$\theta_* = (3, 9/200, 23/100, -11/20, 9/200), \quad c = \frac{1254433}{1250000}. \quad (4.3)$$

Then

$$e_h \in [-.011024692273, .010642371599], \quad |e_h| < .011025 < .025. \quad (4.4)$$

Furthermore,

$$\begin{aligned} p_c &\in [6.508371733, 6.518868974], \\ p_h &\in [6.507844281, 6.519014106]. \end{aligned} \quad (4.5)$$

Proof. Section 5 establishes the common Gaussian structure. Appendices A–C specify all requirements for the validated arithmetic computation, which yields

$$W_P := E_P[(A - cG)^2 I^{-1/2}] \leq 2.172471592802, \quad W_Q \leq 2.242189951600. \quad (4.6)$$

Here $d_\nu = W_\nu / \sqrt{1 - \rho^2}$. The center of the finite linear difference is approximately $-.000202992309237006$, and its complete error radius is at most $.000594238060$. Combining the exact rational endpoints outward with (4.2) gives (4.4). Applying the corresponding error contributions to (3.5) separately for each law gives (4.5). The source code, exact endpoints, and the recomputation of the entire catalog by a second implementation together constitute the executable evidence specified in the appendices. $\square$

4.1 Optimizing the weighted remainder and its width limitation

The common scale c affects the nonlinear conversion bound. Assume, in addition to the structural condition, that

$$m_{AA,\nu} = E_\nu[A^2/\sigma_\nu], \quad m_{AG,\nu} = E_\nu[AG/\sigma_\nu], \quad m_{GG,\nu} = E_\nu[G^2/\sigma_\nu]$$

are finite for $\nu = P, Q$. Put

$$\begin{aligned} a &= m_{AA,P} + m_{AA,Q}, & b &= m_{AG,P} + m_{AG,Q}, \\ d &= m_{GG,P} + m_{GG,Q} > 0. \end{aligned}$$

The nonlinear contribution to the width in (4.2), using exact moment inputs, is

$$\gamma(K_1^{-1} + K_2^{-1})J(c), \quad J(c) = a - 2bc + dc^2.$$

Consequently its unique minimizer on $c > 0$ is

$$c_* = \frac{b}{d}, \quad J(c) = J(c_*) + d(c - c_*)^2. \quad (4.7)$$

The discount and strike factors are positive constants and therefore do not alter this minimizer. This statement concerns the exact nonlinear width; quadrature slack and the linear-transform error allowances can depend on c , so it does not assert that c_* minimizes the final computed enclosure.

In the Heston construction, $\sigma_\nu = \sqrt{1 - \rho^2} \sqrt{I_\nu}$. The common factor $(1 - \rho^2)^{-1/2}$ cancels from (4.7). A numerically preferable representation uses the arithmetic-geometric mean inequality:

$$H_\nu = E_\nu[G^2 I_\nu^{-1/2}], \quad E_\nu^+ = E_\nu[(A - G)G I_\nu^{-1/2}] \geq 0, \quad c_* = 1 + \frac{E_P^+ + E_Q^+}{H_P + H_Q}. \quad (4.8)$$

Both moments use profiles already present in the 91-profile catalog. For example, the Laplace integrand $E_\nu[(A - G)G \exp(-x^2 I_\nu)]$ is nonnegative and nonincreasing. It can therefore be enclosed by the same Darboux construction as (A.5), evaluating the difference of the AG and G^2 profiles at each node before integration. Its continuous tail is bounded by the positive AG tail. At $S_0 = 100$, each positive raw-moment catalog has coefficient sum 10000; the nodal projection allowance for $(A - G)G$ is $20000\delta_Q$. These observations preserve cancellation at the nodal level and avoid subtracting two wide integrated moment enclosures.

If validated positive endpoints give $E_P^+ + E_Q^+ \in [e_-, e_+]$ and $H_P + H_Q \in [d_-, d_+]$, with $d_- > 0$, then

$$c_* \in \left[1 + \frac{e_-}{d_+}, 1 + \frac{e_+}{d_-}\right] = [c_-, c_+].$$

Choosing the rational midpoint $\hat{c} = (c_- + c_+)/2$ gives

$$0 \leq J(\hat{c}) - J(c_*) \leq \frac{d_+(c_+ - c_-)^2}{4\sqrt{1 - \rho^2}}, \quad (4.9)$$

where d_+ encloses the sum of the two $I^{-1/2}$ -weighted G^2 moments, as in (4.8). The selected rational scale must still be substituted into the original square and linear calculations; a ratio enclosure alone is not a new price certificate.

Optimizing c does not recover cancellation between the two models. Even with $Q = P$, an exact linear difference of zero, and exact $d_P = d_Q = D_P(c)$, formula (4.2) has width

$$2\gamma(K_1^{-1} + K_2^{-1})D_P(c).$$

This is positive unless $A = cG$ almost surely. Thus (4.2) is a fixed-step enclosure and, by itself, does not prove that its width tends to zero as h decreases.

For computed inputs the actual width in (4.2) is

$$(u - \ell) + \gamma(d_P + d_Q)(K_1^{-1} + K_2^{-1}). \quad (4.10)$$

The original nonlinear interval has width .020478587753; its linear width is approximately .001188476120, giving displayed total width .021667063872. About 94.5 percent is therefore due to payoff conversion. Increasing precision or reducing an already small projection contribution has limited effect on the final width. Appendix H gives a sufficient coupling inequality for the difference of the remainders. Its additional moments have not been computed at the main point, so it does not improve Corollary 4.2 numerically or assert a Heston convergence rate.

The revised positive-moment calculation at $h = 1/768$ gives

$$c_* \in [1.003427394603466100946724, 1.003669831096000915452372]. \quad (4.11)$$

A midpoint rounded to the 10^{-7} grid selects the rational $5017743/5000000 = 1.0035486$. This is not the exact midpoint of the rational endpoints in (4.11). Formula (4.9) applies to an exact midpoint; for the rounded choice, the distance allowance additionally includes $5 \cdot 10^{-8}$.

Table 3. Effect of the common geometric scale on the nonlinear width.

Common scale	Nonlinear width upper bound at $h = 1/768$
1	.040551110717
1.0035464 (original)	.020478587752
1.0035486 (rounded midpoint)	.020478592980

The original scale is already close to the optimizer. The midpoint-based candidate slightly increases the computed upper width, so the original rational scale is retained. This comparison includes the new moment quadrature, projection, and tail allowances. It does not claim that the exact objective or the complete price certificate has been globally optimized.

5 Recovery of classical conditions and additional models covered

5.1 Deterministic positive integrated variance: $\mathbf{A} \Rightarrow \mathbf{H}$

Define $\mathbf{A}$ as follows: the model's variance coefficient itself is a deterministic function $v(t) \geq 0$, the stock price follows the corresponding linear geometric diffusion, and

$$0 < I_1 = \int_0^{t_1} v(t) dt, \quad \int_0^T v(t) dt < \infty. \quad (5.1)$$

Computability conclusions additionally require these integrals to be supplied as effectively computable inputs.

Take the stock Brownian integral over the first interval as U . Then $\sigma^2 = I_1$ is a positive constant; the remaining log increments and deterministic drift can all be included in $\mathcal{G}$. Every observed price has the common factor in (2.6). The finite positive moments of the joint lognormal distribution give

$$D_\nu(c) = I_1^{-1/2} E_\nu(A - cG)^2 < \infty.$$

The second moment is computed directly from

$$E(A - cG)^2 = \frac{1}{n^2} \sum_{i,j} ES_i S_j - \frac{2c}{n} \sum_i ES_i G + c^2 EG^2 \quad (5.2)$$

whose terms are all Gaussian exponential moments. The corresponding finite transforms and Gaussian tails likewise have explicit bounds. Thus, for a single law with effectively computable inputs, $\mathbf{A} \Rightarrow \mathbf{H}$; for two such laws, Theorem 4.1 further yields $\mathbf{B}$.

This shows that classical lognormal conditioning models fit within the framework. Identifying the common structure allows the same payoff-level argument to apply to different variance mechanisms; the model itself supplies the particular constants and computational catalog.

5.2 Heston and the actual projected kernel: $\mathbf{C} \Rightarrow \mathbf{H}$

Condition $\mathbf{C}$ requires $\kappa, \bar{v}, \xi, v_0 > 0$, $|\rho| < 1$, the model (2.1) or (2.2), observation dates aligned with the discrete grid, and the finite-transform, moment-growth, projection, and infinite-tail inequalities listed in Appendices A--C. The actual verification in this section uses S_0, r, T, h from (2.5) and the parameters in (4.3); the common Gaussian construction below is valid under the structural parameter assumptions just stated.

Condition on the entire variance driver W and on the later independent stock-driver increments $B_t - B_{t_1}$, $t \geq t_1$. The conditioning sigma-field does not include the future Brownian levels B_t , which contain the first-interval value. Let

$$I_P = \int_0^{t_1} V_t dt, \quad I_Q = h \sum_{j=0}^{t_1/h-1} V_j^h.$$

The contribution of B over the first interval is shared by every observation date. Hence

$$U \mid \mathcal{G} \sim N(0, (1 - \rho^2)I_\nu), \quad S_{t_i} = e^U a_i. \quad (5.3)$$

For the continuous model, $v_0 > 0$ and continuity of the variance paths imply $I_P > 0$ almost surely. For the discrete model, there is the exact lower bound

$$I_Q \geq h v_0 > 0. \quad (5.4)$$

The transform and tail bounds in the appendices verify $W_\nu < \infty$ and supply an upper bound, while the exponential-moment growth bounds ensure integrability of the required price terms. Therefore $\mathbf{C} \Rightarrow \mathbf{H}$. At the parameters in (4.3), all these inequalities and finite-node checks hold.

5.3 A strict extension of the class of models covered

Let X_* be the continuous/discrete model pair at (4.3). For the continuous model,

$$\text{Var}(V_t) = \xi^2 \int_0^t e^{-2\kappa(t-s)} E V_s ds > 0 \quad (t > 0), \quad (5.5)$$

so its variance is not a deterministic function. The first step of the discrete model also satisfies

$$Q_h(V_1^h = 0) = \Phi\left(-\frac{(1 - \kappa h)v_0 + \kappa \bar{v} h}{\xi \sqrt{h v_0}}\right) \in (0, 1). \quad (5.6)$$

At the same time, (5.4) still holds. Thus X_* satisfies $\mathbf{C}$ but not $\mathbf{A}$, and the main theorem yields (4.4). The additional model pair has both stochastic variance and a boundary atom induced by projection, while the common Gaussian structure at the payoff level remains valid.

6 A common density perturbation and weak expansions for ten payoffs

For background on weak error expansions and distributional error analysis, see [7, 8, 9]. This section works directly with the differentiability of Gaussian densities. We set $\xi = 0$, retain the original update (2.2), and assume

$$\kappa > 0, \quad \bar{v}, v_0 > 0, \quad t_j = j/12, \quad h = 1/(12m) \leq h_0 < 1/\kappa, \quad m \in \mathbb{N}. \quad (6.1)$$

Let $D = v_0 - \bar{v}$. The continuous and discrete cumulative variances are

$$I(t) = \bar{v}t + \frac{D}{\kappa}(1 - e^{-\kappa t}), \quad I_h(t) = \bar{v}t + \frac{D}{\kappa}[1 - (1 - \kappa h)^{t/h}]. \quad (6.2)$$

The twelve independent components of the log-increment vector Y have distributions

$$N(r/12 - s_j/2, s_j), \quad s_j = I(t_j) - I(t_{j-1}), \quad (6.3)$$

and the discrete law replaces s_j by $s_{h,j}$. Both satisfy the lower bound $s_j, s_{h,j} \geq u_* := \min(v_0, \bar{v})/12 > 0$.

Define

$$a(t) = \frac{D\kappa t e^{-\kappa t}}{2}, \quad B_I(t) = |D|e^{-\kappa t} \left[\frac{\kappa^2 t}{3(1 - \kappa h_0)} + \frac{\kappa^3 t^2}{8(1 - \kappa h_0)^2} \right]. \quad (6.4)$$

Set $d_j = a(t_j) - a(t_{j-1})$ and $b_j = B_I(t_j) - B_I(t_{j-1})$. Appendix D proves that

$$|s_{h,j} - s_j - h d_j| \leq h^2 b_j. \quad (6.5)$$

Theorem 6.1 (A common first-order expansion for bounded payoffs). Under (6.1), let $g(Y)$ be any bounded Borel payoff with range contained in $[l, l + L]$, payable at time τ . Then

$$p_h(g) - p_c(g) = h\beta_g + R_h(g), \quad (6.6)$$

$$\begin{aligned} \beta_g &= e^{-r\tau} E_s \left[g(Y) \sum_j d_j \mathcal{S}_j(Y) \right], \\ |R_h(g)| &\leq e^{-r\tau} \frac{L}{2} C_2 h^2, \end{aligned} \quad (6.7)$$

where

$$\begin{aligned} \mathcal{S}_j &= \frac{G_j^2 - 1}{2s_j} - \frac{G_j}{2\sqrt{s_j}}, \quad J(s) = \frac{1}{2s^2} + \frac{1}{4s}, \quad H_2(s) = \frac{2}{s^2} + \frac{1}{2s}, \\ C_2 &= \sqrt{\sum_j b_j^2 J(u_*)} + \frac{1}{2} \sum_j (|d_j| + h_0 b_j)^2 H_2(u_*). \end{aligned} \quad (6.8)$$

Here G_j denotes the standardized independent normal increments.

Proof. The Gaussian density is twice differentiable in L^1 with respect to its variance parameters. The norm of its first directional derivative is bounded by the corresponding square root of the Fisher information quadratic form, while its second directional derivative satisfies

$$\|Dp_s[z]\|_1 \leq \sqrt{\sum_j z_j^2 J(s_j)}, \quad \|D^2 p_s[z, z]\|_1 \leq \sum_j z_j^2 H_2(s_j).$$

Substituting (6.5) into the L^1 Taylor formula for the density gives $\|p_{s_h} - p_s - hDp_s[d]\|_1 \leq C_2 h^2$. The density remainder integrates to zero, so the payoff can be centered as $g - l - L/2$, yielding (6.7). Appendix D gives the derivative calculations and the full justification by dominated convergence. $\square$

The theorem applies the same leading density perturbation to all ten payoffs. For a put,

$$\beta_{K,\tau} = a(\tau) \frac{S_0 \varphi(d_1)}{2\sqrt{I(\tau)}}, \quad d_1 = \frac{\log(S_0/K) + r\tau + I(\tau)/2}{\sqrt{I(\tau)}}. \quad (6.9)$$

When $v_0 \neq \bar{v}$, all nine put components are nonzero and have the same sign as $v_0 - \bar{v}$. On the family

$$\kappa = 3, \quad \bar{v} = .045, \quad v_0 \in [.03, .06], \quad h_0 = 1/768, \quad (6.10)$$

the second-order remainder for the original Asian payoff has the uniform upper bound .000165217. A separate finite-step density comparison also gives

$$|p_h(\psi) - p_c(\psi)| \leq .004990924. \quad (6.11)$$

The leading Asian coefficient has the exact integral representation (6.7) and an effective absolute bound. This common expansion treats several nonsmooth payoffs through one integrable density perturbation, with every remainder constant retained explicitly.

6.1 Asian coefficient bounds and numerical illustration

The score bound from (6.7) and the first derivative norm gives

$$|\beta_{\text{Asian}}| \leq \frac{15e^{-rT}}{2} \sqrt{\sum_{j=1}^{12} d_j^2 J(s_j)}. \quad (6.12)$$

Table 4 gives pointwise coefficient and remainder bounds for the numerical family (6.10). The latter use (6.5)–(6.8) with each point's lower increment variance. This supplies a common expansion formula and explicit remainder bounds; the signed twelve-dimensional Asian coefficient integral has not been evaluated, and the coefficient enclosures contain zero.

Table 4. Analytic Asian coefficient enclosures and remainder bounds.

Initial variance	Certified coefficient enclosure	Upper bound on $ e_h - h\beta $, $h = 1/768$
0.03	$[-3.441491358, 3.441491358]$	0.000121667773
0.04	$[-0.928189505, 0.928189505]$	0.000036646884
0.05	$[-0.785853082, 0.785853082]$	0.000034532369
0.06	$[-2.057270822, 2.057270822]$	0.000099684585

A Monte Carlo illustration uses 2,097,152 Gaussian increment vectors per point, with common samples across the four step sizes in Table 5. The coefficient estimates are -0.041015 (SE 0.002079), -0.011364 (SE 0.000566), 0.008329 (SE 0.000483), and 0.019917 (SE 0.001274), in increasing order of initial variance. These are sampling estimates, distinct from the analytic enclosures in Table 4.

Table 5. Sampling estimates of the scaled Asian expansion residual.

Initial variance	$h = 1/192$	$1/384$	$1/768$	$1/1536$
0.03	-0.063243	-0.062944	-0.062796	-0.062722
0.04	-0.017226	-0.017145	-0.017105	-0.017085
0.05	0.011698	0.011646	0.011620	0.011607
0.06	0.027912	0.027785	0.027722	0.027690

Entries estimate $(e_h - h\beta_{\text{Asian}})/h^2$ through a Gaussian density-ratio calculation. At $h = 1/768$, their estimated standard errors are .003733, .001000, .000863, and .002319, respectively; estimates across step sizes at a point are correlated. The stable scales are consistent with the second-order expansion. The $h = 1/192$ and $1/384$ columns are diagnostics outside the fixed $h_0 = 1/768$ range used for the numerical constants in (6.10) and Table 4.

7 From a reference family to stochastic-volatility posterior quantiles

7.1 Price transfer for small positive volatility of volatility

Suppose that the deterministic reference variance and its Euler values lie in $[m, M]$, where $0 < m \leq M < \infty$ and $\kappa \geq \kappa_- > 0$, and assume

$$\kappa h \leq 1, \quad 0 \leq \xi \leq \sqrt{\kappa_- M/2}. \quad (7.1)$$

Define

$$C_P = \sqrt{M/(2\kappa_-)}, \quad C_Q = \sqrt{2M/\kappa_-}.$$

The Itô isometry for the continuous model and the 1-Lipschitz property of the discrete positive-part projection give, respectively,

$$E(V_t^\xi - v(t))^2 \leq \xi^2 C_P^2, \quad E(V_j^{\xi,h} - v_j)^2 \leq \xi^2 C_Q^2. \quad (7.2)$$

Combining the deterministic reference lower bound m with Doob's inequality yields

$$E \max |Z_\nu^\xi - Z_\nu^0| \leq \xi C_\nu \left(\frac{T}{2} + 2\sqrt{T/m} \right), \quad \nu = P, Q. \quad (7.3)$$

The original put payoff is K -Lipschitz in log stock price; the original spread payoff is $K_2 = 110$ -Lipschitz in the maximum norm of the fixing log-price vector. Thus (7.3) provides uniform radii from all true calibration prices and the true target price to their reference-family counterparts. Appendix E gives the inductive proof, retaining the effect of projection on the mean.

For example, on the parameter box of positive five-dimensional volume

$$\begin{aligned} \kappa &\in [2.999, 3.001], \quad \bar{v} \in [.04499, .04501], \\ v_0 &\in [.03001, .05999], \quad \xi \in [10^{-5}, 4 \cdot 10^{-5}], \quad \rho \in [-.8, -.3], \end{aligned} \quad (7.4)$$

finite verification gives an absolute bias bound of .020948 for the original Asian and .016702 for each of the nine puts.

7.2 The full joint posterior

The effect of forward numerical error on the likelihood and its normalizing constant is a basic issue in Bayesian approximation [10]. In this section, set $\kappa = 3$, $\bar{v} = .045$, and let

$$u = v_0 \sim U[.03, .06], \quad \xi \sim U[10^{-7}, 10^{-6}], \quad \rho \sim U[-.8, -.3] \quad (7.5)$$

be mutually independent. Write $\zeta = (\xi, \rho)$ and denote its prior probability measure by ν . For model $m = P, Q$, let $c_m(u, \zeta)$ be the true calibration vector and $J_m(u, \zeta)$ the true Asian price. Given synthetic quotes y and a positive-definite covariance matrix Σ_d , define

$$\begin{aligned} w_m(u, \zeta) &= \exp \left[-\frac{1}{2} (c_m - y)^T \Sigma_d^{-1} (c_m - y) \right], \quad \bar{w}_m(u) = \int w_m(u, \zeta) \nu(d\zeta), \\ Z_m &= \frac{1}{.03} \int_{.03}^{.06} \bar{w}_m(u) du, \quad \mu_m(du, d\zeta) = \frac{w_m(u, \zeta)}{.03 Z_m} du \nu(d\zeta). \end{aligned} \quad (7.6)$$

Each model uses its own calibration prices, normalizing constant, and target price.

7.3 A transfer theorem for every quantile level

Theorem 7.1. Suppose that the two marginal posteriors of u admit a coupling (U_P, U_Q) such that $|U_P - U_Q| \leq \delta_u$ almost surely. Assume that the reference targets satisfy

$$|J_P^0(u) - J_P^0(v)| \leq L_0|u - v|, \quad |J_Q^0(u) - J_P^0(u)| \leq \varepsilon_0,$$

and that the true targets satisfy $|J_m(u, \zeta) - J_m^0(u)| \leq e_{m,J}$. Then the left generalized quantiles of the true targets under their respective full joint posteriors satisfy

$$\boxed{|q_Q(p) - q_P(p)| \leq \varepsilon_0 + e_{P,J} + e_{Q,J} + L_0\delta_u \quad (0 < p < 1).} \quad (7.7)$$

Proof. For each model, define the conditional probability kernel

$$K_m(u, d\zeta) = \frac{w_m(u, \zeta)}{\bar{w}_m(u)} \nu(d\zeta).$$

Extend the marginal coupling above by drawing ζ_P and ζ_Q from $K_P(U_P, \cdot)$ and $K_Q(U_Q, \cdot)$, respectively. Fubini's theorem shows that the resulting full marginals are exactly μ_P and μ_Q . The triangle inequality gives

$$|J_Q(U_Q, \zeta_Q) - J_P(U_P, \zeta_P)| \leq \varepsilon_0 + e_{P,J} + e_{Q,J} + L_0\delta_u =: E.$$

Consequently, $F_P(z - E) \leq F_Q(z) \leq F_P(z + E)$. Taking left generalized inverses yields (7.7). The argument also applies to target distributions with atoms. $\square$

7.4 Effective inputs and numerical conclusions

Partition $[.03, .06]$ into 4096 equal intervals. On each interval and over the entire ζ -domain, enclose the true likelihood using (7.3) and the reference Black prices. The same bounds remain valid after integration against the normalized prior ν . Construct enclosures of the marginal CDFs while retaining their shared denominators; an inverse-CDF coupling driven by a common uniform variable then gives δ_u . All constants and the finite pairing algorithm are given in Appendix F.

The single-quote case uses the put with $T = 1, K = 100$, and the nine-quote case uses the full calibration vector. Both use $\Sigma_d = .75^2(.75I_d + .25\mathbf{1}^T)$. The exact quotes are listed in Appendix F. The results are

$$L_0 \leq 213.792201829985, \quad \varepsilon_0 \leq .004493504474, \quad e_{P,J} + e_{Q,J} \leq .000393595478,$$

Case	δ_u	$\sup_{0 < p < 1} q_Q(p) - q_P(p) $
Single quote	3/204800	$\leq .008018821658$
Nine quotes	3/81920	$\leq .012716404217$

(7.8)

The finite partition encloses the continuous posterior and retains the full normalizing mass of each model.

8 Numerical certificates and sensitivity

8.1 From formulas to certified monetary bounds

Table 6 assembles the price difference at the original parameter point. Displayed enclosures and upper bounds are rounded outward. Appendix G gives the exact endpoints, numerical constructions, and reproduction instructions.

Table 6. Error budget for the principal Heston price certificate.

Quantity	Validated upper bound or enclosure
Nonlinear conversion interval	$[-.010227461904, .010251125849]$
Q projection error per linear profile	.000000012711979137
Q projection contribution over all linear profiles	.000004310413
Omitted frequency tail for the continuous model	.000014095712
Omitted frequency tail for the discrete model	.000184373301
Periodization error for one model	.000195729318
Total error bound for the linear difference between the two models	.000594238060
Absolute bound on the true arithmetic Asian bias	.011025

8.2 Step-grid certificates and the width plateau

Table 7 retains the main Heston parameters, payoff, fixing dates, and transform catalogs while changing the Euler step. All successful rows include the full error accounting and alternative numerical checks described in Appendix G.

Table 7. Signed Heston price-difference certificates at four Euler grids.

Step	Euler minus continuous enclosure	Width upper bound
1/192	No certificate: projection guard fails	Not available
1/384	$[-0.014169989568, 0.013839530031]$	0.028009519599
1/768	$[-0.011024692273, 0.010642371599]$	0.021667063872
1/1536	$[-0.010967952032, 0.010766337995]$	0.021734290027

At $h = 1/192$, the fixed projection proof gives $\beta = 184040303/23040000 < 8$, while its recurrence requires $\beta > 8$. This sufficient condition does not certify that mesh. All three successful rows satisfy the absolute error tolerance .025. Table 8 separates their width contributions; its exact unrounded entries sum to the interval widths.

Table 8. Width contributions in price units for the three successful grids.

Width contribution	1/384	1/768	1/1536
Payoff remainder	0.026754977094	0.020478587752	0.020171631177
Linear projection	0.000265787793	0.000008620825	2.65628E-7
P frequency tail	0.000028191423	0.000028191423	0.000028191423
Q frequency tail	0.000177646019	0.000368746602	0.000751284529
Periodization	0.000782917272	0.000782917272	0.000782917272
Rounding and endpoints	1E-12	1E-12	1E-12

Table 8 reports width contributions rather than one-sided error radii; rounding them separately can change their displayed sum. The nonlinear shares are approximately 95.5, 94.5, and 92.8 percent as the grid is refined. At the finest step, the increase in the discrete frequency-tail contribution exceeds the decreases in the payoff remainder and projection contribution. The frequency-tail bound in (C.7) uses the first-step lower bound $i_0 = hv_0$, so refining the Euler grid alone does not tighten this fixed inversion design. This width plateau describes the enclosure, rather than the true price error.

8.3 A second stochastic-variance parameter point

Keep the contract, twelve monthly fixings, $h = 1/768$, and all parameters in (4.3) except the initial variance, which is now $v_0 = .04$. The weighted moments, projection prefixes, transforms, and continuous and discrete tails are evaluated afresh at this point. The same growth estimate gives $\exp(.6 + 4v_0) = \exp(.76)$, so the retained $\exp(.78)$ moment allowance remains conservative. All real-loading, denominator, coefficient, and complex branch conditions pass both numerical constructions.

Table 9. Complete Heston certificate at initial variance .04.

Quantity	Certified enclosure or upper bound
Continuous price	[6.499318797594, 6.509948941256]
Projected-Euler price	[6.498150434416, 6.510702536853]
Euler minus continuous	[−0.011798506840, 0.011383739259]
Signed-interval width	0.023182246099

The enclosure combines the nonlinear remainder with projection, both frequency tails, periodization, and outward arithmetic. Its six exact width contributions and the complete alternative checks are recorded in Appendix G’s package. The wider interval than at $v_0 = .045$ reflects the changed integrated-variance weight and error bounds. This second point demonstrates a separate complete execution; it does not certify a neighborhood or alter the Feller index.

8.4 Coverage and sensitivity design

The main point has Feller index $2\kappa\bar{v}/\xi^2 = 2700/529$, approximately 5.104. Sensitivity to the variance boundary, correlation, strikes, initial variance, and fixing frequency requires new moment, projection, and tail inputs for each case. In particular, changing to 52 or 252 fixings requires an aligned Euler grid; $h = 1/768$ cannot be retained. The first fixing also changes the weight $I_1^{-1/2}$ in the payoff bound. The path comparisons below explore these sensitivities without extending the certified parameter domain.

8.5 Independent path diagnostics for boundary and fixing sensitivity

The path study uses 32,768 paths per case and couples an Euler grid to one with twice as many steps. Table 10 reports three of the fourteen cases, selected by Feller index; the full study also varies correlation, strikes, initial variance, and step size. The coarse-minus-fine estimates have sampling error and retain the finer model’s discretization bias.

Table 10. Coupled path diagnostics at three Feller indices.

Case	Coarse minus fine, estimate and standard error	Projection fraction, coarse
Feller 5.104	-0.0000269; SE 0.0004717	0.0000000
Feller 0.999	0.0001129; SE 0.0012453	0.0011268
Feller 0.639	0.0009133; SE 0.0018375	0.0066991

All three price-difference estimates have magnitude below one estimated standard error, so Table 10 resolves neither their signs nor a systematic increase in pricing bias. The proposal frequencies support only the increased incidence of negative variance proposals near and below the boundary; they are not projection-error bounds.

For 12, 52, and 252 fixings, the sampled means of $I_1^{-1/2}$ are 16.489, 34.078, and 74.871. The aligned coarse grids have 768, 832, and 1008 steps, respectively, with twice as many fine steps. This illustrates the sensitivity of the smoothing weight, while also changing the time grid.

8.6 Continuous-model valuation and numerical risk budgets

Suppose that a valuation system provides a price enclosure $[q_L, q_U]$ for the same discrete model. Theorem 4.1 gives $e_h \in [\ell, u]$, hence the continuous-model price satisfies

$$p_c \in [q_L - u, q_U - \ell]. \quad (8.1)$$

For (4.3), the direct continuous-price interval (4.5) has width below .010498. These two routes provide enclosures in price units for valuation and implementation checks.

8.7 Stability of the price distribution after calibration

A change in numerical implementation affects both the calibration likelihood and the target price. Theorem 7.1 retains both effects: $\varepsilon_0 + e_{P,J} + e_{Q,J}$ controls the target-price difference at fixed parameters, and $L_0\delta_u$ controls the target change caused by movement of the parameter posterior.

On the original 4096-cell grid, the transport contribution accounts for about 39 percent of the single-quote bound and 62 percent of the nine-quote bound. This motivates the grid refinement in the next subsection. A wider nine-quote enclosure reflects its conservative transport contribution and does not imply a larger true numerical bias.

8.8 Posterior grid refinement

Table 11 keeps the same prior (7.5), synthetic observations, likelihood covariance, and target Lipschitz constant. Each row verifies the full-cell likelihood bounds and coupling inequalities and bounds the displacement of every Asian-price quantile.

Table 11. Certified Asian-price quantile displacement under posterior grid refinement.

Cells	Single quote	Nine quotes	Coupling steps, single/nine
4096	0.008018821658	0.012716404217	2 / 5
8192	0.006452960804	0.010367612938	2 / 7
16384	0.006061495591	0.009584682511	3 / 12

The common nonmesh error contributions remain unchanged. Coupling step counts increase with refinement, so the final bound does not scale exactly as the cell width. The new bounds at 16384 cells are approximately 24.4 percent and 24.6 percent smaller than the original single- and nine-quote bounds. Adaptive cells and local target Lipschitz constants are not evaluated in this revision.

The absolute 2.5, 50, and 97.5 percent quantile brackets concern the initial-variance marginal. For the continuous-model posterior with 4096 cells, the single-quote brackets are approximately $[\text{.0307690}, \text{.0307764}]$, $[\text{.0444067}, \text{.0444141}]$, and $[\text{.0590917}, \text{.0590992}]$; the nine-quote brackets are $[\text{.0308203}, \text{.0308350}]$, $[\text{.0435131}, \text{.0435425}]$, and $[\text{.0587182}, \text{.0587256}]$. These brackets describe parameter uncertainty. Absolute Asian-price target quantiles have not been evaluated, so the displacement bounds do not yet measure the change relative to the target’s posterior uncertainty scale.

9 Domain of validity

The structural conditions are sufficient for certification. Section 5 verifies a deterministic-variance class and the specified Heston model and projected kernel; Section 8 varies the step at the principal point and certifies a second initial variance. Extending these enclosures to a parameter neighborhood requires uniform moment, transform, projection, and tail inequalities. The small-positive- ξ box in Section 7 is a separate certified region.

Section 6 gives a common expansion formula and explicit remainder bounds for $\xi = 0$. The nine put coefficients are nonzero when $v_0 \neq \bar{v}$, whereas the Asian coefficient is given by a score integral and analytic bounds containing zero. An expansion for positive ξ requires additional regularity and integrable remainders for the projection kernel and payoff kinks.

The posterior displacement bounds apply to the restricted prior (7.5) and synthetic quotes. Transfer on the five-dimensional parameter domain

$$[2, 4] \times [\text{.03}, \text{.06}] \times [\text{.18}, \text{.28}] \times [-\text{.8}, -\text{.3}] \times [\text{.03}, \text{.06}]$$

requires uniform forward-error and posterior-transport inputs. Absolute Asian-price target quantiles and the signed Asian leading-coefficient integral remain uncomputed.

The separate-law remainder produces the width plateau in Section 8. Proposition H.1 provides a direct coupled-remainder inequality. Its additional coupled moments are certified for a deterministic-variance example in Appendix H, but have not been established for the stochastic-variance Heston examples. Controlling those moments and tightening the discrete frequency tail are distinct mathematical requirements for a shrinking certificate width.

10 Conclusion

The common Gaussian structure connects classical lognormal conditioning with stochastic-variance models discretized by projected Euler. The central lemma converts the kink remainder of the original arithmetic payoff into a second moment weighted by the inverse square root of integrated variance. The main theorem combines this quantity with a finite transform certificate to produce a signed price-difference enclosure. The specified Heston instance gives the complete monetary bound .011025, together with a validated interval for the continuous-model price.

The step-grid study shows that payoff conversion dominates the interval width and that the fixed discrete-tail bound can offset gains from a finer Euler grid. A second stochastic-variance point confirms a complete parameter-specific execution, and the deterministic example in Appendix H gives a strictly narrower coupled nonlinear interval. The common Gaussian expansion and posterior coupling provide additional formula-level and price-unit bounds on their stated parameter families.

Acknowledgments and author's note

The author is responsible for the mathematical scope, attribution, and claims. The version history and numerical materials are recorded in Appendix G.

Appendix A. Finite certification of the weighted second moment

This appendix and Appendices B and C use the rational parameters in (4.3). Write

$$\alpha = \xi^2/2, \quad d = \kappa\bar{v}, \quad s = 1 - \rho^2, \quad t_1 = 1/12.$$

Throughout, c denotes the geometric scaling coefficient in (4.3).

A.1 A nonnegative integral and the loading catalog

For $I > 0$,

$$I^{-1/2} = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-x^2 I} dx.$$

Applying Tonelli's theorem to the nonnegative square gives

$$W_\nu = \frac{2}{\sqrt{\pi}} \int_0^\infty F_\nu(x) dx, \quad F_\nu(x) = E_\nu[(A - cG)^2 e^{-x^2 I}]. \quad (\text{A.1})$$

We have $F_\nu \geq 0$, and the function is nonincreasing. In the expansion (5.2), the 144 ordered terms in A^2 combine into 78 terms; there are also 12 terms of the form $S_i G$ and one G^2 term. Each normalized exponential profile has total real stock loading 2, and the remaining loading in the backward monthly recursion satisfies $p \in [0, 2]$. The sum of the absolute coefficients of the 91 terms is

$$S_0^2(1 + c)^2 = 10000(1 + c)^2. \quad (\text{A.2})$$

A killing rate x^2 is imposed in the first month and is zero in all other months.

A.2 The true continuous flow and the discrete algebraic recursion

On an interval with stock loading q and killing rate λ , write the normalized transform as $\exp(\mathcal{A} + \mathcal{B}v)$. The continuous coefficients satisfy

$$\mathcal{B}' = \alpha\mathcal{B}^2 - (\kappa - \rho\xi q)\mathcal{B} + (q^2 - q)/2 - \lambda, \quad \mathcal{A}' = rq + d\mathcal{B}. \quad (\text{A.3})$$

Completing the square in the one-step Gaussian integral gives the discrete algebraic recursion

$$\begin{aligned} \mathcal{A}_j &= \mathcal{A}_f + h(rq + d\mathcal{B}_f), \\ \mathcal{B}_j &= \mathcal{B}_f + h[\alpha\mathcal{B}_f^2 - (\kappa - \rho\xi q)\mathcal{B}_f + (q^2 - q)/2 - \lambda]. \end{aligned} \quad (\text{A.4})$$

Both updates use the value $\mathcal{B}_f$ before the update. Appendix B bounds the difference between the actual positive-part kernel and (A.4). Fixed observation dates are handled by adding the corresponding stock loadings in reverse chronological order.

For real loadings $p \in [0, 2]$ and $\lambda \geq 0$, the continuous vector field is strictly downward at $\mathcal{B} = 1$. Thus the backward flow from terminal value 0 satisfies $\mathcal{B} \leq 1$, and $\mathcal{A} \leq (2r + d)T$. The uniform integrability established in Appendix B.3 identifies this flow with the true transform.

A.3 Validated Darboux enclosures

Take $\Delta x = 1/2$. For the continuous model, use nodes $x_j = j/2$, $0 \leq j \leq 256$; for the discrete model, use $0 \leq j \leq 128$. If the true nodal value lies in $[f_j^-, f_j^+]$, then

$$\Delta x \sum_{j=1}^M \max(0, f_j^-) \leq \int_0^{M\Delta x} F_\nu(x) dx \leq \Delta x \sum_{j=0}^{M-1} \max(0, f_j^+). \quad (\text{A.5})$$

These are the rectangular-sum bounds for a nonnegative, nonincreasing function. For Q , first multiply (A.2) by the per-profile error bound from Appendix B and enlarge the algebraic nodal enclosure in both directions. Then apply (A.5).

A.4 The discrete infinite tail

Since $I_Q \geq i_0 = hv_0$, for $x \geq X$,

$$F_Q(x) \leq F_Q(X)e^{-i_0(x^2 - X^2)}.$$

Using $x^2 - X^2 \geq 2X(x - X)$ gives

$$\int_X^\infty F_Q(x) dx \leq \frac{\overline{F}_Q(X)}{2Xi_0}, \quad X = 64. \quad (\text{A.6})$$

The upper bound $\overline{F}_Q(X)$ includes the actual projection error and rounding.

A.5 The continuous infinite tail

Let $\kappa' = \kappa - 2\rho\xi$, and define

$$\Delta(x) = \sqrt{\kappa'^2 + 4\alpha(x^2 - 1)}, \quad r_1 = \frac{\kappa' - \Delta}{2\alpha}, \quad r_2 = \frac{\kappa' + \Delta}{2\alpha}.$$

The bound $\mathcal{B} \leq 1$ in subsequent months allows each positive-loading profile to be bounded above by taking terminal value 1 for the first-month flow. For $x \geq X = 128$,

$$g = \frac{1 - r_1}{1 - r_2} \in (-1, 0).$$

The cross-ratio solution of the Riccati equation yields

$$\mathcal{B}(t_1) \leq r_1 + \frac{\Delta}{\alpha} e^{-\Delta t_1}, \quad \int_0^{t_1} \mathcal{B}(t) dt \leq r_1 t_1 + \frac{\log 2}{\alpha}. \quad (\text{A.7})$$

Furthermore, $\Delta(x) \geq 2\sqrt{\alpha}(x - 1/X)$ and $\Delta(X)t_1 > 1$, so $\Delta e^{-\Delta t_1}$ is decreasing throughout this tail region. Define

$$C_0 = \frac{\kappa'}{2\alpha} + \frac{1}{X\sqrt{\alpha}}, \quad C_1 = C_0 + \frac{\Delta(X)e^{-\Delta(X)t_1}}{\alpha},$$

$$C = \exp\left\{2r + d + 2rt_1 + d\left(t_1 C_0 + \frac{\log 2}{\alpha}\right) + v_0 C_1\right\}, \quad \zeta = \frac{dt_1 + v_0}{\sqrt{\alpha}}.$$

Each normalized real transform is therefore at most $Ce^{-\zeta x}$. Since $(A - cG)^2 \leq 2A^2 + 2c^2G^2$,

$$\int_X^\infty F_P(x) dx \leq \frac{20000(1 + c^2)C}{\zeta} e^{-\zeta X}. \quad (\text{A.8})$$

All root relations, signs, and transcendental functions are checked using outward-rounded arithmetic.

Combining (A.5)–(A.8) and multiplying by $2/\sqrt{\pi}$ gives

$$\begin{aligned} W_P &\leq 2.172471592801589415440352, \\ W_Q &\leq 2.242189951599049647468543. \end{aligned} \quad (\text{A.9})$$

These moment bounds also establish the integrability required in (2.7).

Appendix B. The actual projection kernel and affine identification

B.1 The one-step projection correction

Let the current variance be $v \geq 0$, and let p be the real part of the stock loading. After the Gaussian tilt induced by the stock loading, the candidate next-step variance satisfies

$$Y \sim N(\eta_p v + dh, \xi^2 h v), \quad \eta_p = 1 - (\kappa - \rho\xi p)h,$$

with the external factor $\exp(rph + \gamma_p hv)$, where $\gamma_p = (p^2 - p)/2$. For a future complex variance coefficient b , if the candidate is $Y = -z < 0$, then

$$|e^{bY^+} - e^{bY}| \leq |b|ze^{|b|z} \leq \frac{|b|h}{\lambda e} \exp[(\lambda/h + |b|)z]. \quad (\text{B.1})$$

Set $s_b = \lambda + h|b|$. The Gaussian moment formula reduces the exponent in the state-dependent error bound to

$$-\frac{v}{h}(s_b \eta_p - \alpha s_b^2 - \gamma_p h^2) - s_b d. \quad (\text{B.2})$$

Take $\lambda = 12$. Suppose p lies in the specified loading range, $\eta_p \geq \eta_*$, $\gamma_p \leq \gamma_*$, and $|b| \leq b_*$. Checking lower bounds for the concave quadratic at the two endpoints $s_b = 12$ and $s_b = 12 + b_* h$ suffices to ensure that the expression in parentheses is at least 8.

B.2 From local error bounds to true prefixes

Define

$$u_0 = 8, \quad u_{j+1} = \eta_* u_j - \alpha u_j^2 - \gamma_* h^2. \quad (\text{B.3})$$

Nonnegativity $u_j \geq 0$ is checked at all 768 levels. For $u \geq 0$, $e^{-uY^+/h} \leq e^{-uY/h}$. Hence the actual stock-weighted prefix Laplace expectations can be bounded above by repeated application of Gaussian moment bounds. This yields the projection error bound for each complete profile:

$$\delta_Q = \frac{b_*}{12e} \exp(C_f + C_{\text{pre}} - 12d) h \sum_{j=0}^{767} \exp \left[-d \sum_{k < j} u_k - \frac{v_0 u_j}{h} \right]. \quad (\text{B.4})$$

Here C_f bounds the future constant term and C_{pre} bounds the stock drift over the true prefix. The two parameter sets are as follows.

Use	Range of p	b_*	η_*	γ_*	$C_f + C_{\text{pre}}$
Weighted second moment	$[0, 2]$	512	$1 - (\kappa - 2\rho\xi)h$	1	$4r + d$
Thirteen linear transforms	$[-1/2, 1]$	1024	$1 - (\kappa - \rho\xi)h$	3/8	$2r + d$

The finite catalog checks both $\text{Re } b \leq 1$ and the modulus bound for the future coefficients. In the first case, (B.4) is multiplied by $10000(1 + c)^2$ to obtain the error allowance at each Laplace node. In the second case, it is multiplied by the sum of the absolute values of all Fourier coefficients. No additional factor of h is applied to this sum.

B.3 Moment growth and removal of stopping

For real stock loading p , take e^{4V} as the variance component of the test function. The coefficient of V in the continuous generator calculation is

$$16\alpha - 4\kappa + 4\rho\xi p + \gamma_p. \quad (\text{B.5})$$

This coefficient is strictly negative for all required loadings, including their $5/4$ -scaled range $p \in [-5/4, 5/2]$. For Q , the positive coefficient 4 allows the direct use of $1 - e^{-4z} \leq 4z$. Together with the Gaussian tail bound at $\lambda = 12$, the additional growth rate caused by projection is at most

$$\frac{4e^{-12d}}{12e}.$$

Over the same enlarged range, the state coefficient in the tilted exponent remains nonpositive. For example, using the worst-case values $\eta_{5/2}$ and $\gamma_* = 15/8$, exact evaluation gives

$$12\eta_{5/2} - 144\alpha - \frac{15}{8}h^2 = \frac{8001336523}{983040000} > 0.$$

The total constant growth satisfies

$$4d + \frac{5}{2}r + \frac{4e^{-12d}}{12e} < .589267620943 < .6. \quad (\text{B.6})$$

Thus the required exponential moments of the true continuous and discrete models admit uniform finite upper bounds. In particular, the normalized moments for the original loading range $p \in [-1, 2]$ are bounded by

$$M_2 = e^{4v_0+.6} = e^{.78} \quad (\text{B.7})$$

First stop the continuous affine expression on bounded variance domains. Raising its modulus to the power $5/4$ scales the stock loadings into the range above. The future positive variance coefficient is at most $5/4 < 4$, and the nonpositive killing contribution can be omitted in an upper bound. Equations (B.5)–(B.6) therefore give a uniform $L^{5/4}$ bound for the stopped family, implying uniform integrability. Removing the stopping identifies the Riccati expression with the true transform expectation. The finite discrete telescoping sum involves only finitely many integrable conditional expectations, to which the same moment-growth estimate applies.

Appendix C. Complete error control for inversion of the thirteen linear transforms

C.1 Damped inversion and the finite sum

For a positive measure μ , write $\hat{\mu}(z) = \int e^{zy} \mu(dy)$. Take

$$\begin{aligned} \delta &= 1/2, & \Delta u &= 1/5, & N &= 640, \\ U &= N\Delta u = 128, & L &= 2\pi/\Delta u = 10\pi. \end{aligned}$$

When the negative moment is finite, $e^{-\delta y}F_\mu(y)$ is integrable and has Fourier transform $\hat{\mu}(-\delta - iu)/(\delta + iu)$. Consequently,

$$F_\mu(y) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{(\delta+iu)y}}{\delta + iu} \hat{\mu}(-\delta - iu) du. \quad (\text{C.1})$$

Substitute (3.6), and set $z_n = \delta + in\Delta u$, $w_0 = 1/2$, and $w_n = 1$ for $n \geq 1$. The coefficients in the finite real sum are

$$\begin{aligned} a_{0,n} &= \frac{e^{-r} \Delta u w_n}{\pi z_n} (95e^{z_n y_{95}} - 110e^{z_n y_{110}}), \\ a_{i,n} &= \frac{e^{-r} \Delta u w_n}{\pi z_n} \frac{100}{12} (e^{z_n y_{110}} - e^{z_n y_{95}}), \quad 1 \leq i \leq 12. \end{aligned} \quad (\text{C.2})$$

The corresponding transforms are $Ee^{-z_n Y}$ and $Ee^{Z_i - z_n Y}$. The zero frequency receives half weight, while the highest retained frequency receives full weight. The real part of the remaining loading in a stock profile lies in $[-1/2, 1]$; its squared modulus requires real loadings in $[-1, 2]$.

C.2 Periodization error

Poisson periodization gives

$$\sum_{k \in \mathbb{Z}} e^{-\delta k L} F_\mu(y + kL).$$

For $k > 0$, use the total mass $M_{\mu,0}$. For $k = -j < 0$, use $F_\mu(y - jL) \leq e^{y-jL} M_{\mu,-1}$. The sum of the nonzero-index terms is then at most

$$\frac{M_{\mu,0}}{e^{\delta L} - 1} + \frac{M_{\mu,-1} e^y}{e^{(1-\delta)L} - 1}. \quad (\text{C.3})$$

Here $M_{0,0} = 1$, $M_{A,0} \leq 100e^{.01}$, $M_{0,-1} \leq e^{.78}$, and $M_{A,-1} \leq 100e^{.78}$. Taking absolute values term by term in the price combination gives the periodization error allowance for a single law:

$$E_{\text{alias}} = e^{-.01} \frac{205 + 200e^{.01} + e^{.78}(195e^{y_{95}} + 210e^{y_{110}})}{e^{5\pi} - 1}. \quad (\text{C.4})$$

The difference between the two models is charged $2E_{\text{alias}}$. The frequency tails below ensure absolute summability of the Fourier coefficients, while (C.3) ensures absolute and uniform convergence of the periodization on compact intervals. Fourier uniqueness therefore identifies both representations with the same continuous function.

C.3 Frequency decay from the common Gaussian factor

Under the conditioning in (5.3), the total imaginary loading of each profile on the first-month factor U is $-u$. Set $K = E[e^{\text{real loading}} | \mathcal{G}]$. Conditional Gaussian integration, followed by Cauchy--Schwarz and conditional Jensen, gives

$$|\Phi_\nu(u)| \leq E[Ke^{-u^2 s I/2}] \leq \sqrt{Ee^{2 \text{real loading}}} \sqrt{Ee^{-su^2 I}} \leq \sqrt{M_2 L_\nu(u)}. \quad (\text{C.5})$$

Here $L_\nu(u) = E_\nu e^{-su^2 I}$. For the average of the twelve terms defining F_A , retain the additional price-level factor 100. Define

$$C_* = e^{-.01} (195e^{\delta y_{95}} + 210e^{\delta y_{110}}).$$

Under Q , $I \geq i_0 = hv_0$, so

$$L_Q(u) \leq L_Q(U) e^{-si_0(u^2 - U^2)} \quad (u \geq U).$$

A validated upper bound for the true $L_Q(U)$ requires only 64 backward steps. Starting from $b = A = 0$, use

$$A \leftarrow A + dhhb, \quad b \leftarrow b + h(\alpha b^2 - \kappa b - sU^2). \quad (\text{C.6})$$

At each level, verify $b \leq 0$. The inequality $e^{bY^+} \leq e^{bY}$ then ensures that this Gaussian recursion is an upper bound for the actual positive-part kernel. At the end, $\bar{L}_Q(U) = e^{A+bv_0}$. Bounding the discrete sum of a positive decreasing envelope by its integral gives

$$E_{\text{tail},Q} \leq \frac{C_* \sqrt{M_2 \bar{L}_Q(U)}}{\pi s i_0 U^2}. \quad (\text{C.7})$$

The integral inequality used here is

$$\int_U^\infty e^{-a(u^2 - U^2)} \frac{du}{u} \leq \frac{1}{2aU^2}.$$

For P , define

$$\Delta_U = \sqrt{\kappa^2 + 4\alpha s U^2}, \quad C_0 = \kappa/(2\alpha), \quad C_1 = C_0 + \Delta_U e^{-\Delta_U t_1}/\alpha,$$

$$C_P = \exp\{d(t_1 C_0 + \log 2/\alpha) + v_0 C_1\}, \quad \gamma = \frac{1}{2} \sqrt{s/\alpha} (dt_1 + v_0).$$

The same CIR cross-ratio bound as in (A.7), now with terminal value $b = 0$, gives $L_P(u) \leq C_P e^{-2\gamma u}$ for every $u \geq U$. Hence

$$E_{\text{tail},P} \leq \frac{C_* \sqrt{M_2 C_P}}{\pi \gamma U} e^{-\gamma U}. \quad (\text{C.8})$$

C.4 Branches of the continuous complex flow

Write $\kappa_q = \kappa - \rho\xi q$, $\gamma_q = (q^2 - q)/2$, and $D_q = \sqrt{\kappa_q^2 - 4\alpha\gamma_q}$. Divide each month into eight subintervals of length $\Delta t = 1/96$. The denominator of the Möbius flow is

$$\mathcal{D}_t = \cosh(D_q t/2) + (\kappa_q - 2\alpha b) \frac{\sinh(D_q t/2)}{D_q}.$$

The integral of the flow is

$$\int_0^{\Delta t} b(t) dt = \frac{\kappa_q \Delta t/2 - \log \mathcal{D}_{\Delta t}}{\alpha}. \quad (\text{C.9})$$

On every subinterval, verify

$$\cosh(|D_q| \Delta t/2) - 1 + \frac{|\kappa_q - 2\alpha b| \Delta t}{2} \cosh(|D_q| \Delta t/2) < 1. \quad (\text{C.10})$$

Absolute-value bounds on the power series imply $|\mathcal{D}_t - 1| < 1$ along the entire subinterval. The principal logarithm therefore remains continuously connected to the initial value 1 of the denominator. At the boundary $\text{Re } b = 1$, the imaginary-part quadratic form

$$-\alpha(\text{Im } b)^2 - \rho\xi(\text{Im } q)(\text{Im } b) - (\text{Im } q)^2/2$$

is nonpositive and the remaining real drift is strictly negative. Thus the continuous flow satisfies $\text{Re } b \leq 1$. Appendix B.3 identifies the affine expression with the true transform.

C.5 Combination and precision

Let $\hat{e}_L$ be the finite proxy difference. The true linear difference is enclosed by

$$\hat{e}_L + [-E_L, E_L], \quad E_L = 2E_{\text{alias}} + E_{\text{tail},P} + E_{\text{tail},Q} + \delta_Q \sum_{n,i} |a_{i,n}|. \quad (\text{C.11})$$

All finite sums and coefficients are evaluated as Arb enclosures. Substituting the exact rational endpoints gives

$$E_L \leq .000594238059839992574505.$$

Combining this with the asymmetric payoff remainder obtained from (A.9), according to Theorem 4.1, yields (4.4).

Appendix D. Gaussian density expansions and finite-step bounds

D.1 Variance expansion

For $t/h \in \mathbb{N}$, set

$$x = -\frac{t}{h} \log(1 - \kappa h) - \kappa t \geq 0.$$

The power series for the logarithm gives

$$0 \leq x - \frac{\kappa^2 th}{2} \leq \frac{\kappa^3 th^2}{3(1 - \kappa h)}, \quad x \leq \frac{\kappa^2 th}{2(1 - \kappa h)}.$$

Using $|1 - e^{-x} - x| \leq x^2/2$ and substituting

$$I_h(t) - I(t) = \frac{D}{\kappa} e^{-\kappa t} (1 - e^{-x})$$

yields $|I_h(t) - I(t) - ha(t)| \leq h^2 B_I(t)$. Adding the two cumulative remainder bounds at adjacent times gives (6.5).

D.2 Density derivatives

For a single increment, let $X = Y - r/12$. Its density is

$$f_s(y) = (2\pi s)^{-1/2} \exp[-X^2/(2s) - X/2 - s/8].$$

Hence

$$\partial_s \log f_s = -\frac{1}{2s} + \frac{X^2}{2s^2} - \frac{1}{8}, \quad \partial_s^2 \log f_s = \frac{1}{2s^2} - \frac{X^2}{s^3}.$$

With $X = -s/2 + \sqrt{s}G$, the first-order score has mean zero and second moment $J(s) = 1/(2s^2) + 1/(4s)$. Moreover,

$$E|\partial_s^2 \log f_s| \leq \frac{3}{2s^2} + \frac{1}{4s}.$$

Independence of the coordinates makes the cross-score expectations vanish, yielding the two L^1 derivative bounds used in the proof of Theorem 6.1. On a variance box with a positive lower bound and a finite upper bound, the first and second density derivatives are uniformly dominated by $C(1 + |y|^4)e^{-c|y|^2}$. Difference quotients, parameter integrals, and payoff integrals may therefore be interchanged as required.

D.3 A complete finite-step comparison

For two variance vectors $s, \tilde{s}$, suppose that their endpoints and the segment joining them satisfy $s_j, \tilde{s}_j \geq u_j > 0$. Integrating the first-order score along the segment gives

$$\text{TV}(p_s, p_{\tilde{s}}) \leq \left\{ \sum_j (\tilde{s}_j - s_j)^2 \left(\frac{1}{8u_j^2} + \frac{1}{16u_j} \right) \right\}^{1/2}. \quad (\text{D.1})$$

For a payoff with range length L , the price difference is at most $Le^{-r\tau} \text{TV}$. On (6.10), using the uniform bounds $|D| \leq .015$ and $u_j \geq .03/12$, finite computation gives

$$|e_{h,\text{Asian}}| \leq .004990923469, \quad |R_{h,\text{Asian}}| \leq .000165216416.$$

The posterior module uses the tighter actual minimum variance for each month, so its reference bound ε_0 at fixed parameters can be smaller than this uniform bound.

D.4 An effectively computable representation of the leading coefficient

For rational parameters and the explicit Lipschitz payoffs considered here, the leading coefficient in (6.7) is an effectively computable integral. Let

$$S_d(G) = \sum_j d_j \left[\frac{G_j^2 - 1}{2s_j} - \frac{G_j}{2\sqrt{s_j}} \right], \quad F = \sum_j d_j^2 J(s_j).$$

After centering the payoff, omitting the region $\max_j |G_j| > R$ incurs error at most

$$e^{-r\tau} \frac{L}{2} \sqrt{24F} e^{-R^2/4}. \quad (\text{D.2})$$

This follows directly from Cauchy--Schwarz and a union bound for Gaussian tails. On $[-R, R]^{12}$, define

$$M_S = \sum_j |d_j| \left[\frac{R^2 + 1}{2s_j} + \frac{R}{2\sqrt{s_j}} \right],$$

$$L_S = \sum_j |d_j| \left[\frac{R}{s_j} + \frac{1}{2\sqrt{s_j}} \right].$$

If the payoff has Lipschitz constant L_z as a function of the fixing log-price vector, the integrand has Lipschitz constant at most

$$L_H = L_z \left(\sum_j \sqrt{s_j} \right) M_S + \frac{L}{2} L_S.$$

The sum of box-midpoint values weighted by exact Gaussian box probabilities, for boxes of side length ℓ , has total interior error at most $e^{-r\tau} L_H \ell / 2$. Choosing R, ℓ together with (D.2) therefore gives a finite algorithm with at most $\lceil 2R/\ell \rceil^{12}$ boxes. This representation establishes computability; Section 9 distinguishes this representation from an actually executed high-precision evaluation of the leading coefficient.

Appendix E. Perturbation constants for positive volatility of volatility

In the continuous model, $D_t = V_t^\xi - v(t)$ admits a linear variation-of-constants representation. Since $EV_t^\xi = v(t) \leq M$, the Itô isometry gives

$$ED_t^2 = \xi^2 \int_0^t e^{-2\kappa(t-s)} EV_s^\xi ds \leq \frac{\xi^2 M}{2\kappa_-}. \quad (\text{E.1})$$

The discrete deterministic reference remains nonnegative. The positive-part map is 1-Lipschitz, and the Gaussian innovation has conditional mean zero. Thus, writing $D_j^{(2)} = E(V_j^\xi - v_j)^2$, we obtain

$$D_{j+1}^{(2)} \leq (1 - \kappa h)^2 D_j^{(2)} + \xi^2 h EV_j^\xi \leq (1 - \kappa h)^2 D_j^{(2)} + \xi^2 h (M + \sqrt{D_j^{(2)}}). \quad (\text{E.2})$$

Take $C_Q^2 = 2M/\kappa_-$. Condition (7.1) gives $\xi C_Q \leq M$ and $1 - (1 - \kappa h)^2 \geq \kappa_- h$. Induction from the zero initial error then yields $D_j^{(2)} \leq \xi^2 C_Q^2$. This step retains the effect of projection on EV_j^ξ .

For the deterministic reference $v \geq m$,

$$|\sqrt{x} - \sqrt{v}|^2 = \frac{|x - v|^2}{(\sqrt{x} + \sqrt{v})^2} \leq |x - v|^2/m.$$

The drift contribution to the log-stock difference is bounded by $T\xi C_\nu/2$. The Itô isometry and Doob's L^2 inequality bound the martingale contribution by $2\xi C_\nu \sqrt{T/m}$, proving (7.3). Correlation between the stock Brownian motion and the variance driver does not alter these inequalities for adapted stochastic integrals.

For a put, the absolute derivative with respect to log stock price is at most K on the active region. For the capped Asian, consider a line segment in the fixing log-price vector. On the active region $K_1 < A < K_2$, the gradient has ℓ^1 -norm $A \leq K_2$; elsewhere it is zero. Piecewise absolute continuity gives the global Lipschitz constant 110. Thus, for maturity T ,

$$\begin{aligned} e_{\nu,i} &= K_i e^{-rT_i} \xi_{\max} C_\nu \left(T_i/2 + 2\sqrt{T_i/m} \right), \\ e_{\nu,J} &= 110 e^{-r} \xi_{\max} C_\nu (1/2 + 2/\sqrt{m}). \end{aligned} \tag{E.3}$$

In (7.5), $m = .03$, $M = .06$, $\kappa = 3$, so $C_P = .1$, $C_Q = .2$, giving

$$e_{P,J} \leq .000131198492451391, \quad e_{Q,J} \leq .000262396984902781. \tag{E.4}$$

For (7.4), first enclose the reference variance difference over 256 closed intervals in κ using outward arithmetic. Then apply

$$|p_h^\xi - p_c^\xi| \leq |p_h^0 - p_c^0| + 110\xi(C_P + C_Q)(1/2 + 2/\sqrt{m})$$

to obtain the full-box price bound stated in the main text. The remaining parameters are enclosed through uniform bounds on $m, M, |v_0 - \bar{v}|$, rather than by substituting endpoint samples for full enclosures.

Appendix F. Posterior inputs, normalization, and finite coupling

F.1 Exact quotes

The quotes are ordered by $T = 1/4, 1/2, 1$, with $K = 90, 100, 110$ within each maturity. The following rational numbers are taken as the given synthetic inputs:

$$y = \begin{pmatrix} 526706580970763/562949953421312 \\ 285901443526503/70368744177664 \\ 754049251176359/70368744177664 \\ 580752698287259/281474976710656 \\ 794307614955649/140737488355328 \\ 830842010134081/70368744177664 \\ 269207942741719/70368744177664 \\ 137257851354697/17592186044416 \\ 1918003280297167/140737488355328 \end{pmatrix}. \quad (\text{F.1})$$

These fractions are the exact binary64 values in the reference input file; the single-quote case uses the eighth entry. They are inputs to the likelihood, and the auxiliary computation that produced them need not be regarded as a proof of continuous-model prices.

F.2 Reference prices and target constants

Let

$$b_P(t) = \frac{1 - e^{-3t}}{3}, \quad b_Q(t) = \frac{1 - (1 - 3h)^{t/h}}{3},$$

$$I_\nu(T; u) = .045T + (u - .045)b_\nu(T).$$

The reference put price is computed by the exact Black formula

$$p_\nu(K, T; u) = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1),$$

$$d_1 = \frac{\log(S_0/K) + rT + I_\nu/2}{\sqrt{I_\nu}}, \quad d_2 = d_1 - \sqrt{I_\nu}. \quad (\text{F.2})$$

Since $b_\nu(T) > 0$ and vega is positive, the reference price over each entire u -cell is enclosed by its endpoint values. Expanding both endpoints by (E.3) then encloses the true prices over the entire cell and all ζ .

Let

$$a_{\nu,j} = b_\nu(t_j) - b_\nu(t_{j-1}),$$

$$z_{\nu,j}(u) = .045/12 + (u - .045)a_{\nu,j}, \quad B_* = 15e^{-.01}.$$

The centered score bound in Appendix D gives

$$L_0 = \frac{B_*}{2} \sqrt{\sum_j a_{P,j}^2 J(z_{P,j}(.03))},$$

$$\begin{aligned}\varepsilon_0 &= \frac{B_*}{2} \sqrt{\sum_j \Delta_j^2 J(\underline{z}_j)}, \\ \Delta_j &= .015 |a_{Q,j} - a_{P,j}|, \quad \underline{z}_j = \min(z_{P,j}(.03), z_{Q,j}(.03)).\end{aligned}\tag{F.3}$$

F.3 Likelihood enclosures over entire cells

For $d = 1$ or 9 ,

$$\Sigma_d^{-1} = \frac{64}{27} \left(I_d - \frac{\mathbf{1}\mathbf{1}^T}{d+3} \right).$$

Writing the residual vector as r , its potential is a sum of nonnegative terms:

$$\frac{1}{2} r^T \Sigma_d^{-1} r = \frac{32}{27(d+3)} \left(3 \sum_i r_i^2 + \sum_{i < j} (r_i - r_j)^2 \right).\tag{F.4}$$

Evaluate this expression and its exponential outward, then round to the exact integer lattice with spacing 2^{-96} , to obtain

$$0 < l_{\nu,i} \leq w_{\nu}(u, \zeta) \leq u_{\nu,i} \leq 1$$

throughout the entire cell and nuisance-parameter domain. Integration against the probability measure $\nu(d\zeta)$ gives the same enclosure for $\bar{w}_{\nu}(u)$.

F.4 Shared normalization and inverse-CDF pairing

Each cell has the same prior mass. At a node x_k , denote the sums of lower and upper mass bounds to its left by $L_<, U_<$, and those to its right by $L_>, U_>$. The marginal CDF G_{ν} satisfies

$$\underline{G}_{\nu}(x_k) = \frac{L_<}{L_< + U_>}, \quad \overline{G}_{\nu}(x_k) = \frac{U_<}{U_< + L_>}.\tag{F.5}$$

The function $A/(A+B)$ increases in A and decreases in B , so these enclosures retain the shared mass in the numerator and denominator. The normalizing constant is also enclosed directly by summing all weights.

Let U be a common uniform random variable and set $U_{\nu} = G_{\nu}^{-1}(U)$. If $U_{\nu} \in [x_i, x_{i+1}]$, then

$$U \in [\underline{G}_{\nu}(x_i), \overline{G}_{\nu}(x_{i+1})].\tag{F.6}$$

For each P-cell, enumerate the leftmost and rightmost indices j_-, j_+ of all Q-cell intervals whose corresponding ranges in (F.6) can intersect its range. Set

$$\delta_u = \Delta x \max_i \max\{|i - (j_+ + 1)|, |i + 1 - j_-|\}, \quad \Delta x = .03/4096.\tag{F.7}$$

The true cell pair must lie in this candidate set; hence $|U_P - U_Q| \leq \delta_u$ almost surely. All comparisons use exact integers and fractions.

The enclosures of the normalizing constants for the true three-dimensional prior, obtained from all cells, are

Case	Z_P	Z_Q
Single quote	[.935743002637584884, .935913179433726711]	[.935667223967444023, .935927287640792653]
Nine quotes	[.620128921272339539, .621155228251382142]	[.619743550733119263, .621256330676942526]

The target radii from Appendix E, together with (F.3) and (F.7), yield (7.8). This procedure encloses each model’s own full posterior.

Appendix G. Reproducible implementation

The public repository is <https://github.com/130U/certified-valuation-arithmetic-asian-options>. The immutable versions have different roles. The original numerical kernels and reference payloads are frozen at 5bc72cf6036fdd73fea9c8bc6c85f85fd474b79a. The first major-revision package, including the step-grid and posterior-grid experiments, is available at commit cf0d242f1590c7ca07ab8ca7afb57da5a583032b. Checking out only the older kernel baseline does not retrieve those experiments. The root REVISION-MANIFEST.json records the source, input, output and receipt byte identities for each published package; code/MANIFEST.json retains the original numerical package identity.

Module	Corresponding result
asian	Theorem 4.1 and Corollary 4.2 at the original point.
tt	Theorem 6.1 and constants on the deterministic face (6.10).
small-xi	The five-dimensional perturbation box (7.4).
posterior-zero	Gaussian-reference posterior and target constants.
posterior	The restricted prior (7.5) and Theorem 7.1.

The reference runtime is unoptimized CPython 3.12 on Windows x86-64 with python-flint==0.8.0. From a checkout containing the revision, the commands below create the environment, verify the original package and execute the original Asian certificate with both implementations. Each double hyphen is a literal command-line character.

```
py -3.12 -m venv .venv
./venv/Scripts/python.exe -m pip install -r code/requirements.txt
./venv/Scripts/python.exe code/run.py verify
./venv/Scripts/python.exe code/run.py run --module asian --independent
```

The run prints its new directory under code/runs/. Replace run-ID below with that directory name. To execute every original module, replace asian in the preceding command with all.

```
../venv/Scripts/python.exe code/run.py check --run code/runs/run-ID
```

Installation may require network access. The numerical calculations do not. No dependency wheel is included; offline installation requires an authentic compatible wheel obtained separately. Assertions enforce mathematical conditions and resource contracts. Each worker uses one thread and a 256 MiB hard memory limit.

The original completed execution and the revision experiments are preserved in `code/revision/results/evidence.zip`. Its inventory binds each member's bytes and SHA-256. The index `code/revision/results/INDEX.md` gives archive extraction and ledger-replay commands; `code/revision/README.md` gives fresh calculation commands and experiment inputs. Saved-result verification and fresh kernel execution are distinct operations.

The first-revision ledger `code/revision/results/revision-summary.json` maps each printed numerical row to the exact source field, result, hash and receipt. The original weighted moments are in `code/reference/asian-remainder-result.json`; the linear center, transform contributions and signed price enclosure are in `code/reference/asian-linear-result.json`. Exact synthetic observations are in `code/data/synthetic_quotes.json`. Appendices A, B and C describe the nonlinear catalog, original projected recurrence and linear inversion, respectively.

The reference comparison includes every numerical JSON field, including arrays and rational endpoints. Only hash metadata and six named elapsed-time fields are excluded. The alternative constructions described below share Arb and the analytical inequalities; numerical agreement verifies the implemented finite calculations within those assumptions, without supplying an independent derivation of the common inequalities.

The experiment archive retains stopped attempts and successful completions. At 192 steps the fixed projection inequality fails its required exponent bound. The initial 384-step attempt stopped at the original numerical radius target; the successful enclosure records that target as false while preserving every analytical condition. At 1536 steps the initial worker exceeded its original wall-time allowance. The completed run multiplies that finite allowance by $\text{ceil}(N/768)$, retaining the mathematical guards, precision, thread and memory limits. Separate weighted-moment experiments use the analogous declared runtime scaling. Receipts distinguish analytical exclusion, numerical acceptance targets and resource stops. Reported task times are machine- and load-specific.

G.1 Numerical constructions and recorded cost

The nonlinear-moment module evaluates 24,388 continuous real-flow blocks and 815,360 discrete real-recursion steps at the principal point. The linear module evaluates 641 frequencies and thirteen profiles, for 6,399,744 discrete steps and 799,968 continuous branch checks. The principal programs use 256-bit Arb [11]; the check programs use 384-bit Arb with different numerical constructions.

For the weighted remainder, `check-asian-remainder.py` replaces the principal root/cross-ratio Riccati formula by a hyperbolic two-by-two flow, constructs the loading catalog from ordered pairs, and separately indexes the reverse discrete grid. It accumulates the projection prefix as a product rather than exponentiating a sum. For linear inversion, `check-asian-linear.py` replaces the principal real/imaginary discrete recursion by a complex `acb` recursion and expresses the continuous hyperbolic blocks through exponentials. It verifies every frequency contribution and recomputes the finite error allowances without

calling the principal numerical functions. The checks share Arb, the analytic identities, and the theoretical bounds. They test implementation consistency and arithmetic stability rather than supplying independent derivations of those bounds. The root-flow and matrix-flow calculations also enclose the same grouped positive moments used in (4.11).

Historical worker times for the original nonlinear and linear modules were approximately 12.02 and 33.20 seconds, respectively, with one thread and a 256 MiB hard memory limit per process. In the step-grid revision, the summed outer wall times for the five Asian tasks were approximately 122.3, 184.0, and 431.6 seconds at 384, 768, and 1536 steps. These latter observations include controllers and check programs under the recorded execution load; they are not portable performance guarantees. Each receipt retains its job times, operation counts, and resource contract.

G.2 Sampling illustrations and posterior marginal envelopes

The Asian coefficient illustration uses binary64 arithmetic and 2,097,152 Gaussian increment vectors per point. It evaluates the exact deterministic-variance Gaussian coefficient integrand and density-ratio residual with common vectors across the four step sizes. Subtracting the known zero-mean score and using `expm1` reduces cancellation in the scaled residual. Inputs, seeds, formulas, sampling estimates, and estimated standard errors are recorded in `code/revision/results/asian-beta-diagnostic.json`. Its normal-approximation sampling intervals are distinct from the analytic coefficient enclosures and deterministic remainder bounds in Section 6.

The separate NumPy path study uses 32,768 paths for each of fourteen cases, coupling the original projected Euler grid to a grid with twice as many steps. The file `code/revision/results/path-diagnostics.json` records seeds, binary64 inputs, sampled prices, standard errors, and negative-proposal counts. Its fine law retains discretization bias, so these sampling results do not enclose the continuous-model price.

The posterior grid summary, `code/revision/results/posterior-grid-summary.json`, retains full rational CDF envelopes and the 2.5, 50, and 97.5 percent quantile brackets of the initial-variance marginal. The absolute Asian-price target quantiles are not part of that computation.

G.3 Second-point and coupled-moment reproduction

The completed second Heston point in Section 8.3 and the executed deterministic coupling example in Appendix H are frozen, together with their programs, inputs, results and receipts, at commit 1959f8f078e1fedd590000d6166cb20429ff31db. This immutable data revision contains the earlier revision experiments as well. From its root and the same Windows environment, use a new directory name:

```
./venv/Scripts/python.exe code/revision/round2/asian_parameter_point.py prepare --directory new-point
./venv/Scripts/python.exe code/revision/round2/asian_parameter_point.py run --directory new-point
```

The run writes its result capsule in the new directory. The point configuration changes only v_0 to $1/25$. The preparation checks the original numerical manifest and constructs explicit parameter copies. The weighted, linear, pilot and two alternative-construction jobs recompute all affected quantities. Declared worker budgets are 120 seconds for each weighted calculation, 360 for each linear calculation and 30 for the pilot; the one-thread and 256 MiB limits remain fixed. The recorded five-job outer time is approximately 97.5

seconds. The archive `code/revision/round2/results/evidence-point-v004.zip` contains all 52 execution members and their inventory. Its SHA-256 and standard-library replay instructions are in `code/revision/round2/README.md`. The exact six-component width ledger is in `code/revision/round2/results/point-v004-result.json`.

The deterministic coupling example uses finite closed Gaussian moments at 384 and 512 bits. Its fresh calculation command is

```
./venv/Scripts/python.exe code/revision/round2_coupling_example.py --output coupling.json
```

The exact result is `code/revision/round2/results/coupling-v004-result.json`. The self-contained proof and source/result hash receipt are in `docs/revision-round2/`. A separate scalar covariance-moment evaluation also agrees with the enclosures. This comparison verifies the nonlinear component on $\xi = 0$; it does not replace the positive- ξ pricing calculation.

G.4 Version history

The research and original manuscript originate in 2024. The numerical package was consolidated and verified in 2026. The first major revision and this targeted revision are dated 7 October 2026. The baseline, first-revision and second-point data commits above identify these different reproducibility scopes.

Appendix H. A direct coupled-remainder inequality

The following sufficient inequality preserves cancellation between the two conditional remainders under a coupling of their conditional variables.

Proposition H.1 (A remainder bound under a coupling). Fix $K > 0$ and $c > 0$, assume that both marginal laws satisfy $H_s(c)$, so that the remainders are finite, and couple conditional triples (a_P, b_P, σ_P) , (a_Q, b_Q, σ_Q) , where $b_\nu = c g_\nu$, all entries are positive, and each marginal is the conditional triple in (2.6). Set

$$s_- = \min(\sigma_P, \sigma_Q), \quad s_+ = \max(\sigma_P, \sigma_Q), \quad z = \max(|a_P - b_P|, |a_Q - b_Q|),$$

$$M_K = \frac{e^{2s_+^2}}{K s_- \sqrt{2\pi}}, \quad N_K = \frac{e^{2s_+^2}(1 + 4/e + 8s_+^2)}{K s_-^2 \sqrt{2\pi}},$$

$$\Xi_K = M_K z (|a_Q - a_P| + |b_Q - b_P|) + \frac{1}{2} N_K z^2 |\sigma_Q - \sigma_P|.$$

If $E\Xi_K \leq B_K < \infty$, then

$$|R_{K,Q} - R_{K,P}| \leq B_K. \tag{H.1}$$

In particular, writing $B = e^{-rT}(B_{K_1} + B_{K_2})$, the exact nonlinear contribution to the call-spread price difference belongs to $[-B, B]$. This interval may be intersected with the nonlinear interval in (4.2) before adding the linear-transform interval. Under the diagonal coupling for identical conditional laws, $\Xi_K = 0$.

Proof. Let Z be standard normal and define

$$F_K(x; s) = E[(x e^{sZ} - K)^+], \quad r_K(a, b, s) = F_K(a; s) - F_K(b; s) - (a - b) \partial_x F_K(b; s).$$

Conditional integration gives $R_{K,\nu} = Er_K(a_\nu, b_\nu, \sigma_\nu)$. For $x, s > 0$, with $y = \log(x/K)$,

$$H_K(x, s) := \partial_x^2 F_K(x; s) = \frac{K}{sx^2\sqrt{2\pi}} e^{-y^2/(2s^2)}, \quad \partial_s H_K = \frac{H_K}{s} \left(\frac{y^2}{s^2} - 1 \right).$$

Completing the square gives

$$H_K(x, s) = \frac{e^{2s^2}}{Ks\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{y}{s} + 2s \right)^2 \right].$$

For $w = y/s + 2s$,

$$|(w - 2s)^2 - 1| e^{-w^2/2} \leq (2w^2 + 8s^2 + 1) e^{-w^2/2} \leq 4/e + 8s^2 + 1.$$

Therefore, whenever $s \in [s_-, s_+]$, $0 \leq H_K \leq M_K$ and $|\partial_s H_K| \leq N_K$, uniformly in $x > 0$. Differentiation of the remainder and Taylor's integral formula give

$$\begin{aligned} \partial_a r_K &= \partial_x F_K(a; s) - \partial_x F_K(b; s), & \partial_b r_K &= -(a - b) H_K(b, s), \\ \partial_s r_K &= (a - b)^2 \int_0^1 (1 - t) \partial_s H_K(b + t(a - b), s) dt. \end{aligned}$$

Interpolate the three entries linearly between the coupled endpoints. All interpolated a, b, s stay positive, $|a - b| \leq z$, and $s \in [s_-, s_+]$. The fundamental theorem of calculus and the displayed derivative bounds yield, for each coupled pair,

$$|r_K(a_Q, b_Q, \sigma_Q) - r_K(a_P, b_P, \sigma_P)| \leq \Xi_K.$$

Taking expectations proves (H.1). Applying the triangle inequality to the two strikes and multiplying by the discount proves the price-unit statement. $\square$

The s_-^{-2} factor requires coupled-moment control beyond (2.7) and the marginal affine catalog. If a family of verified couplings satisfies $E\Xi_K \rightarrow 0$, then (H.1) gives vanishing remainder differences. No such convergence bound or rate for the positive-volatility-of-volatility Heston example is asserted here.

H.1 An executed coupling example on the deterministic-variance face

At $\xi = 0, \kappa = 3, \bar{v} = 9/200, v_0 = 1/25, S_0 = 100, r = 1/100, T = 1, h = 1/768$, retain the twelve monthly fixings, the call spread with strikes 95 and 110, and $c = 1254433/1250000$. Write $s_{\nu,j} = I_\nu(t_j) - I_\nu(t_{j-1})$, using the exact continuous and Euler cumulative variances in (6.2). Couple the laws by twelve common independent standard normals. With $\sigma_\nu = \sqrt{s_{\nu,1}}$, remove the first centered normal factor and put

$$\begin{aligned} m_{\nu,i} &= rt_i - I_\nu(t_i)/2, & \ell_{\nu,i,j} &= \sqrt{s_{\nu,j}} \mathbf{1}_{\{j \leq i\}}, & 2 \leq j \leq 12, \\ a_{\nu,i} &= 100 e^{m_{\nu,i} + \ell_{\nu,i} \cdot Z}, & a_\nu &= \frac{1}{12} \sum_i a_{\nu,i}, & g_\nu &= 100 e^{\bar{m}_\nu + \bar{\ell}_\nu \cdot Z}, & b_\nu &= cg_\nu. \end{aligned}$$

Here $Z = (Z_2, \dots, Z_{12})$, and bars denote averaging over fixing index $i = 1, \dots, 12$. The drift includes the first-month term $-s_{\nu,1}/2$; hence the removed factor has the centered Gaussian law required by $H_s(c)$. Here the stock driver $\rho W + \sqrt{1 - \rho^2} B$ is combined into a

single Brownian motion. On $\xi = 0$ the variance is deterministic and the price law is independent of ρ ; hence $\sigma_\nu^2 = s_{\nu,1}$, using the full first-month variance. The separate-law comparison below uses this same conditioning.

All conditional second moments, including mixed P/Q moments, follow from

$$E[100e^{m+u \cdot Z} 100e^{n+v \cdot Z}] = 10000e^{m+n+\|u+v\|^2/2}.$$

Define

$$q_\nu = E(a_\nu - b_\nu)^2, \quad Z_2 = q_P + q_Q, \quad D_a = E(a_Q - a_P)^2, \quad D_b = E(b_Q - b_P)^2.$$

For the deterministic M_K, N_K in the coupling proposition, the pointwise bound $z^2 \leq (a_P - b_P)^2 + (a_Q - b_Q)^2$ and Cauchy--Schwarz give the executable upper bound

$$E\Xi_K \leq B_K := M_K \sqrt{Z_2} (\sqrt{D_a} + \sqrt{D_b}) + \frac{1}{2} N_K Z_2 |\sigma_Q - \sigma_P|.$$

The finite Gaussian-moment computation at 384 and 512 bits yields

$$E\Xi_{95} \leq .000023887739528734380066, \quad E\Xi_{110} \leq .000020630320502088782784.$$

Consequently the discounted nonlinear spread contribution belongs to the symmetric interval with radius $e^{-r}(B_{95} + B_{110})$, whose width is at most .000088150195864703911227. For the same laws, conditioning, and scale, the separate-law estimate has width in

$$[.015383637151510699801154, .015383637151510699801155].$$

The ratio of widths is below .005731. Strict improvement is checked using the outward upper endpoint of the coupled width and the lower endpoint of the separate width. The comparison uses $D_\nu = e^{2\sigma_\nu^2} q_\nu / \sigma_\nu$, so all weighted moments refer to the same Gaussian factor.

This is a closed-moment enclosure of the coupled remainder expectations, with no Monte Carlo or unbounded-domain quadrature. It establishes narrowing of the nonlinear component at one deterministic-variance point. It does not compute a complete price interval, a signed Asian leading coefficient, or a coupled certificate for $\xi > 0$. The two precisions and two algebraic Gaussian-moment expressions share the analytical proof and Arb; they are arithmetic consistency checks.

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